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Workshop 1 - Sentiment Analysis of Food Preferences for Event-Based Menu Recommendation Using NLP

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Contents

1	Executive Summary	2
2	Project Definition	2
2.1	Topic Angle	2
2.2	Context and Motivation	2
2.3	Goal	3
2.4	Scope and Limitations	3
3	Source Characterization	3
3.1	API Source — Spoonacular	3
3.2	Web Scraping Source — Eater NY (RSS Feed)	4
4	Data Samples and Source Validation Evidence	4
4.1	Spoonacular API Response Sample	5
4.2	Eater NY Scraped Content Sample	5
5	Extraction Strategy	6
5.1	Ingestion Periodicity	6
5.2	Raw Data Format	7
5.3	Metadata Fields	7
6	Functional User Definition	7
7	User Stories	8
8	Pipeline Architecture Overview	8
8.1	Layer Descriptions	8
8.2	Orchestration Strategy	9
9	Results & Discussion	10
10	Repository	11
11	Conclusions	11

1 Executive Summary

Topic angle: This report defines an NLP-driven data pipeline for sentiment analysis of food preferences in event-based menu recommendation. The project focuses on New York City, where restaurants, hotels, and catering teams operate in a competitive market shaped by culinary trends, dietary restrictions, and event-specific guest expectations.

Problem context: Event planners need to select dishes that fit the occasion, audience profile, and current food trends. However, menu decisions are often based on intuition or isolated customer feedback. Public culinary content, recipe metadata, and food-related opinions provide a stronger foundation for identifying which dishes are likely to be well received in specific event contexts.

Proposed solution: The system combines structured data from the Spoonacular API with unstructured editorial content from the Eater NY RSS feed. The Bronze layer stores raw JSON files, the Silver layer applies cleaning and NLP preprocessing, and the Gold layer is designed to aggregate sentiment and trend signals for dashboard consumption.

Expected outcome: The pipeline produces traceable data products that support menu recommendation by linking dishes, ingredients, event types, dietary constraints, and sentiment signals. This report documents the topic angle, source characterization, source validation samples, extraction strategy, functional user, user stories, and pipeline architecture overview required for the first workshop.

2 Project Definition

2.1 Topic Angle

The project topic is **sentiment analysis of food preferences for event-based menu recommendation**. The central idea is to transform food-related public opinion and structured recipe metadata into actionable insights for catering managers and event planners.

2.2 Context and Motivation

New York City stands as one of the most gastronomically diverse urban environments in the world, hosting thousands of restaurants and luxury hotels that regularly organize events ranging from corporate dinners to weddings and cultural celebrations. For catering managers and event planners, the ability to anticipate what dishes will resonate with specific audiences is strategically and commercially critical.

Public opinion expressed through online platforms—particularly reviews, food blogs, and culinary forums—constitutes an underutilized source of insight for menu design. Structured recipe data from APIs like Spoonacular, when combined with unstructured

opinion data, enables a recommendation layer that goes beyond generic popularity rankings.

2.3 Goal

The goal is to design a data engineering and NLP pipeline capable of collecting, processing, and analyzing food-related opinions to support evidence-based menu recommendation for event planning scenarios.

2.4 Scope and Limitations

- **Scope:** Integration of API and web data, sentiment analysis, trend extraction, and event-oriented recommendation support.
- **Limitations:** API rate limits, noisy user-generated or editorial content, possible missing metadata, and limited contextual understanding of highly specific event preferences.

3 Source Characterization

3.1 API Source — Spoonacular

Spoonacular is a food and recipe API that provides structured data about recipes, ingredients, and nutritional information. It was selected as the primary structured data source because it offers direct access to culinary metadata that forms the catalog of dishes the recommendation system evaluates.

Characteristics:

- REST API with JSON responses; authentication via API key.
- Endpoint used: `/recipes/findByIngredients`, queried through the wrapper function `collect_recipes_by_ingredient`, which accepts the following parameters:
 - `ingredient` — main ingredient to search for.
 - `number` — number of recipes to retrieve (default: 10).
 - `intolerances` — dietary intolerances to filter out (e.g., *gluten*, *dairy*).
 - `diet` — dietary preference filter (e.g., *vegan*, *vegetarian*).
- Returns structured records including recipe title, matched ingredients, missing ingredients, and recipe ID for further lookup.

Relevance: This API provides the structured ingredient and recipe layer of the pipeline — the “what a dish is” side of the recommendation engine. The `intolerances` and `diet` parameters are particularly relevant for event-based menu planning, as they allow filtering dishes to match the dietary needs of specific guest profiles directly at ingestion time.

Validation: A successful request returns a JSON array where each object includes `id`, `title`, `usedIngredientCount`, `missedIngredientCount`, and `missedIngredients`. Listing 1 provides a raw-style validation sample used to confirm the expected API structure.

3.2 Web Scraping Source — Eater NY (RSS Feed)

Eater NY is a leading food blog and news publication dedicated to the New York City culinary scene. Initially, platforms like TripAdvisor were considered; however, strict Web Application Firewalls (WAF) prevented ethical, automated extraction. Consequently, the Eater NY RSS feed was selected as the primary unstructured data source because it provides highly reliable, opinion-rich, and public-driven insights into emerging food trends and popular dishes in NYC without triggering anti-bot mechanisms.

Characteristics:

- **Opinion-rich text:** Article summaries and descriptions contain descriptive text about trending dishes, ingredients, and culinary expectations — essential for NLP, sentiment analysis, and entity extraction.
- **Reliable Delivery (XML/RSS):** Utilizing the native RSS feed bypasses aggressive DOM changes and WAF restrictions common in standard HTML scraping, ensuring a stable and ethical ingestion process.
- **Structured Metadata:** Articles are accompanied by pre-assigned categories (e.g., "NYC Restaurant Openings", "NYC Restaurant News") that allow targeted filtering of event-relevant dining trends.
- **Fields to extract:** article title, URL, author, article summary (raw text), categories, publication date, and extraction timestamp.

Relevance: Eater NY complements Spoonacular by providing the “market demand” layer — identifying what dishes or ingredients are currently trending in the NYC food scene. Together, both sources enable the pipeline to bridge unstructured public interest with structured recipe metadata, forming the basis of the event-driven recommendation engine.

Validation: The RSS extraction process returns semi-structured article records containing a title, URL, author, summary text, categories, publication date, and extraction timestamp. Listing 2 shows the scraped-content structure used to validate this source.

4 Data Samples and Source Validation Evidence

This section provides JSON snippets used as evidence that both sources can be collected and stored in the Bronze layer. The snippets are intentionally kept short while preserving the fields required for validation, traceability, and later NLP processing.

4.1 Spoonacular API Response Sample

Listing 1 shows a representative successful response from the Spoonacular recipe endpoint. The sample confirms that the API returns recipe identifiers, titles, ingredient matches, missing ingredients, and a count of likes, which are sufficient to build the structured dish catalog.

Listing 1: Spoonacular API validation response sample.

```
1  [
2    {
3      "id": 73420,
4      "title": "Apple Or Peach Strudel",
5      "image": "https://spoonacular.com/recipeImages/73420-312x231.jpg",
6      "imageType": "jpg",
7      "usedIngredientCount": 1,
8      "missedIngredientCount": 3,
9      "missedIngredients": [
10     {
11       "id": 18371,
12       "amount": 1.0,
13       "unit": "package",
14       "name": "phyllo dough",
15       "original": "1 package phyllo dough"
16     }
17   ],
18   "usedIngredients": [
19     {
20       "id": 9003,
21       "amount": 6.0,
22       "unit": "large",
23       "name": "apples",
24       "original": "6 large baking apples"
25     }
26   ],
27   "unusedIngredients": [],
28   "likes": 1
29 }
30 ]
```

4.2 Eater NY Scraped Content Sample

Listing 2 shows the JSON structure produced by parsing the Eater NY RSS feed. The sample validates that the web source provides opinion-rich article summaries and meta-data that can be cleaned and transformed into NLP-ready text.

Listing 2: Eater NY RSS scraped-content validation sample.

```
1  [  
2    {  
3      "article_title": "NYC Restaurant Openings and Dining Trends",  
4      "article_url": "https://ny.eater.com/rss/index.xml",  
5      "article_id": "eaterny-rss-validation-sample",  
6      "author": "Eater NY",  
7      "article_summary": "A roundup of restaurant news, openings, chef  
        updates, and dishes drawing attention across New York City dining  
        .",  
8      "categories": ["NYC Restaurant News", "NYC Restaurant Openings"],  
9      "location_focus": "New York City",  
10     "published_date": "2026-03-01T00:00:00Z",  
11     "date_extracted": "2026-03-01T00:00:00Z"  
12   }  
13 ]
```

5 Extraction Strategy

5.1 Ingestion Periodicity

Both sources follow a **daily schedule**, with an additional **event-driven trigger** activated whenever a new event is registered in the system (e.g., a corporate dinner, wedding, or private gathering).

- **Spoonacular API:** Daily. Recipe catalogs and ingredient metadata are collected once per day to keep the Bronze layer synchronized with current recipe availability while still controlling API quota consumption. When an event is registered, an additional on-demand run is triggered with the **diet** and **intolerances** parameters adjusted to match the guest profile of that event.
- **Eater NY (RSS Feed):** Daily. Culinary trends, restaurant news, and public opinions are refreshed every day so the NLP layer can capture recent sentiment shifts relevant to menu planning. Event triggers also apply here to refresh the latest gastronomic news corpus before a specific event.

Periodicity justification: Daily ingestion provides a consistent refresh cycle for both structured recipe metadata and unstructured food-trend text, ensuring that menu recommendations reflect recent culinary signals without waiting for a longer batch window. Weekly ingestion balances data freshness with resource efficiency. The event-driven layer ensures that when a planner needs up-to-date recommendations for a specific occasion, the pipeline can be re-run on demand without altering its schedule.

5.2 Raw Data Format

All ingested data is persisted as **JSON files** with datetime-based naming to enable version control and historical traceability:

- **Spoonacular:**
spoonacular_recipes_YYYYMMDD_HHMMSS.json
- **Eater NY (RSS Feed):**
eaterny_trends_YYYYMMDD_HHMMSS.json

Files are stored in `datalake_bronze/` without any transformation applied at ingestion time.

5.3 Metadata Fields

The following metadata fields are persisted alongside the raw text content of each record to ensure data quality and traceability:

- **source** — origin platform (Spoonacular).
- **timestamp** — datetime of ingestion (YYYY-MM-DD HH:MM:SS).
- **author** — reviewer username or recipe author.
- **date** — original publication date of the record.
- **url** — source URL for traceability and re-validation.
- **language** — detected language of the text content.
- **category** — cuisine type or restaurant category.
- **event_keywords** — event-related terms detected in the record (e.g., *wedding*, *corporate*, *catering*).

6 Functional User Definition

The primary functional user is a **Hotel or Restaurant Catering Manager / Event Planner** operating in New York City.

Profile:

- Medium technical knowledge—comfortable with web dashboards but not with raw code.
- Requires actionable insights: top-rated dishes per event type, trending ingredients, dietary demand signals.
- Makes weekly or per-event decisions on menu curation and event proposals.
- Operates in a competitive market where guest satisfaction is a primary KPI.

Justification: This user benefits from the pipeline by replacing ad-hoc intuition with data-backed menu choices. Visualizing which dishes receive the highest positive sentiment for weddings, corporate events, or private parties enables confident, client-facing proposals aligned with current NYC dining trends.

7 User Stories

User stories are prioritized by importance for dash-board design:

1. **[High]** As a catering manager, I want to see the highest-rated dishes for a specific type of event so I can offer clients a competitive menu.
2. **[High]** As an event planner, I want to filter dish suggestions by dietary restriction (vegan, gluten-free, halal) so that I can accommodate all guest profiles in a single proposal.
3. **[High]** As a restaurant manager, I want to monitor sentiment trends over time for specific dish categories so that I can detect which cuisines are gaining or losing popularity in NYC.
4. **[Medium]** As a data analyst, I want to view data quality metrics (missing fields, ingestion success rate, record counts) so that I can ensure the pipeline produces reliable inputs for the recommendation engine.
5. **[Medium]** As an event planner, I want to identify dishes frequently mentioned alongside specific NYC neighborhoods so that I can tailor menus to location-specific audiences.
6. **[Low]** As a manager, I want to receive automated alerts when a dish category experiences a sudden sentiment drop so that I can proactively remove it from upcoming event menus.
7. **[Low]** As a data analyst, I want to compare sentiment distributions between API-sourced recipe metadata and reviews so that I can validate the consistency of the recommendation model.

8 Pipeline Architecture Overview

The system follows the **Medallion Architecture**, organizing data into three progressive layers, as shown in Figure 1.

8.1 Layer Descriptions

A. Bronze — Raw Ingestion

Two **Apache Airflow** DAGs handle data collection: one queries the Spoonacular API via Python `requests`, and a second ingests data from the web source to be defined. Airflow was chosen for its retry logic, failure alerts, and monitoring interface. All responses are stored as timestamped JSON files in `datalake_bronze/` without transformation.

B. Silver — Cleaning and NLP

An Airflow **FileSensor** detects new Bronze files and triggers the Silver DAG automatically. **Pandas** handles data cleaning and normalization. The NLP pipeline applies entity extraction, sentiment scoring via **VADER** — suited for short-form opinion text — and keyword and topic extraction. Output is stored as **Parquet** in `datalake_silver/`, offering significant compression and faster analytical reads compared to JSON.

C. Gold — Aggregations

Pandas computes daily summaries: top dishes by sentiment per event type and trend windows. **PySpark** is available as a scale-out option if data volume grows beyond memory limits.

D. Dashboard

Plotly Dash reads the Gold summaries and renders an interactive dashboard accessible via browser. All components are containerized with **Docker** and launched via a single `docker-compose.yml`.

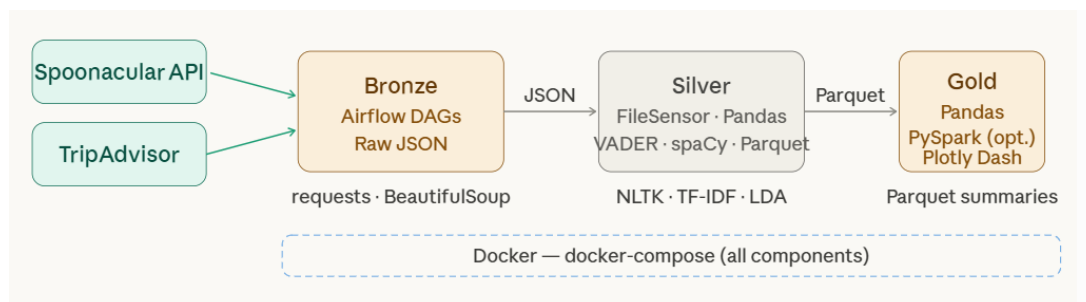


Figure 1: Medallion architecture pipeline showing the Bronze, Silver, and Gold data layers.

8.2 Orchestration Strategy

The pipeline is orchestrated using **Apache Airflow**, a platform that allows defining workflows as **DAGs** (Directed Acyclic Graphs). A DAG is a Python script that specifies a set of tasks and the order in which they must execute — each task only runs after the previous one completes successfully, and the flow never loops back. Airflow provides a web interface to monitor task status, inspect logs, and configure automatic retries on failure.

The pipeline uses two DAGs:

1. **DAG 1 — Bronze ingestion** (scheduled daily at 00:00 UTC). This DAG runs on a fixed schedule and executes three tasks in sequence: call the Spoonacular API, call the web data source, and persist both responses as timestamped JSON files in `datalake_bronze/`. If any task fails, Airflow retries it automatically and sends an alert.
2. **DAG 2 — Silver processing** (event-driven). Instead of running on a fixed schedule, this DAG starts with a `FileSensor` — a special Airflow task that monitors `datalake_bronze/` and waits until new files appear. The moment DAG 1 writes new data, the sensor triggers the rest of the pipeline automatically: data cleaning with Pandas, NLP processing (entity extraction, sentiment scoring, topic modelling), and finally writing the results as Parquet files to `datalake_silver/`.

Once DAG 2 completes, a **Gold aggregation task** runs downstream: it reads the Silver Parquet files, computes the daily summaries (top dishes by sentiment, trend windows, dietary frequencies), and writes the final outputs to `datalake_gold/`. These aggregated files are immediately consumed by the Plotly Dash dashboard.

9 Results & Discussion

Initial validation confirms that both sources successfully provide sufficient data for sentiment and entity analysis. The Spoonacular API offers highly structured culinary data, while the Eater NY RSS feed provides rich, unstructured textual opinions on market trends. However, a preliminary data quality assessment revealed specific preprocessing challenges that must be addressed in the Silver layer pipeline:

- **Eater NY (Scraping Noise):** The extracted article summaries contain significant structural noise. The raw text includes embedded HTML entities (e.g., `’`, ` `), journalistic boilerplate, and photo credits (e.g., “Photo by Patrick McMullan via Getty Images”). This unstructured data will require aggressive cleaning and text normalization using regular expressions before NLP models can accurately process the sentiment.
- **Spoonacular API (Nested Structures):** While the API data is clean, it is heavily nested and contains operational metadata unnecessary for NLP. The transformation process will require flattening nested JSON arrays (such as iterating through `missedIngredients` to extract only the `name` or `originalName` fields), standardizing complex ingredient strings (e.g., parsing “1/2 stick (4 Tbs) unsalted butter, cut into cubes” into core entities), and dropping non-textual fields like image URLs and numerical IDs to optimize storage and processing speed.

Note: The JSON snippets in Listings 1 and 2 demonstrate the raw structures used to validate both ingestion sources.

10 Repository

The full project source code, data samples, and documentation are available at the following GitHub repository:

<https://github.com/Salda1308/Food-Gastronomy-Trends-Sentiment>

11 Conclusions

This workshop establishes the conceptual and technical foundation for an NLP-driven menu recommendation pipeline targeting event planners and catering managers in New York City. Two complementary data sources were identified and validated: the **Spoonacular API**, which provides structured recipe and ingredient metadata, which supplies opinion-rich user reviews from NYC dining establishments.

The combination of both sources enables the pipeline to bridge the gap between “what a dish is” (structured API data) and “how people feel about it” (unstructured review text), forming a solid basis for event-aware menu recommendations.

The proposed Medallion architecture (Bronze $\rightarrow$ Silver $\rightarrow$ Gold) ensures a clean separation between raw ingestion, NLP processing, and aggregated insights, while Apache Airflow provides reliable orchestration through daily and event-driven DAG triggers.

Future work will focus on implementing the Bronze DAGs, completing the 20-record data samples from both sources, developing the Silver NLP pipeline (entity extraction, VADER sentiment scoring, TF-IDF keyword extraction), and deploying the initial Plotly Dash prototype.

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